

Lesson 3: Linear dependence and independence

Introduction to linear dependence

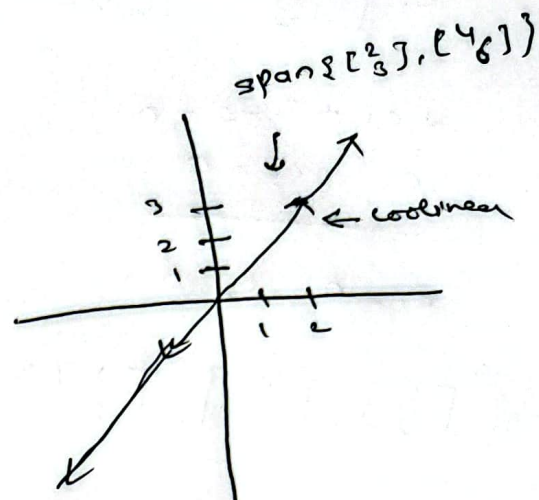
$\left\{ \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 6 \end{bmatrix} \right\} \leftarrow$ linearly dependent

$$c_1 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + c_2 \begin{bmatrix} 4 \\ 6 \end{bmatrix}$$

$$c_1 \begin{bmatrix} 2 \\ 3 \end{bmatrix} + c_2 2 \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$(c_1 + 2c_2) \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$c_3 \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$



$\begin{bmatrix} 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 7 \\ 2 \end{bmatrix}, \begin{bmatrix} 9 \\ 5 \end{bmatrix} \rightarrow$ linearly dependent
 $v_1 + v_2 = v_3$

$$\text{span}(v_1, v_2) = \mathbb{R}^2$$

$\begin{bmatrix} 7 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \end{bmatrix} \leftarrow$ linearly independent
 v_1, v_2

$$\text{span}(v_1, v_2, v_3) = \mathbb{R}^3$$

$$\text{span}(v_1, v_2) = \mathbb{R}^2$$

$\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 7 \end{bmatrix} \rightarrow$ linearly independent
 $= \mathbb{R}^3$

More on linear independence

Linear dependence

$$S = \{v_1, v_2, \dots, v_n\}$$

Linearly dependent iff

$$v_1 = a_2 v_2 + a_3 v_3 + \dots + a_n v_n$$

$$0 = \underbrace{1}_{\text{non zero}} v_1 + a_2 v_2 + a_3 v_3 + \dots + a_n v_n$$

$$c_1 v_1 + c_2 v_2 + \dots + c_n v_n = 0 = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$\Rightarrow$ for some c_i 's not all are zero at least 1 is non zero

Assume $c_1 \neq 0$

$$v_1 + \frac{c_2}{c_1} v_2 + \dots + \frac{c_n}{c_1} v_n = 0$$

$$\frac{c_2}{c_1} v_2 + \dots + \frac{c_n}{c_1} v_n = -v_1$$

$\left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \end{bmatrix} \right\} \rightarrow$ linearly independent set of vectors.

$$c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ 2 \end{bmatrix} = 0$$

c_1 or c_2 non zero $\Rightarrow$ linearly dependent

c_1 & c_2 both zero $\Rightarrow$ linearly independent set

$$2c_1 + 3c_2 = 0$$

$$c_1 + 2c_2 = 0$$

$$c_1 = 0, c_2 = 0$$

$\left\{ \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\} \leftarrow$ linearly independent

$$c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 3 \\ 2 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$2c_1 + 3c_2 + c_3 = 0$$

$$c_1 + 2c_2 + 2c_3 = 0$$

$$\underline{\underline{c_3 = -1}}$$

$$2c_1 + 3c_2 - 1 = 0$$

$$2c_1 + 2c_2 + 2 = 0$$

$$\underline{\underline{c_1 = -4, c_2 = 3}}$$

$$-4 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 3 \begin{bmatrix} 3 \\ 2 \end{bmatrix} - 1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

satisfies $\Rightarrow$ linearly dependent.

span and linear independence example

$\left\{ \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} \right\} \subset \mathbb{R}^3$ spans $\mathbb{R}^3$?
linearly independent?

$$c_1 \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} + c_3 \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$c_1 + 2c_2 - c_3 = a$$

$$-c_1 + c_2 = b$$

$$2c_1 + 3c_2 + 2c_3 = c$$

$$c_3 = \frac{1}{11} (3c - 5a + b)$$

$$c_2 = \frac{1}{3} (b + a + c_3)$$

$$c_1 = a - 2c_2 + c_3$$

$$c_1 + 2c_2 - c_3 = a$$

$$3c_2 - c_3 = b \quad | \times a \quad 3c_2 = b + a + c_3$$

$$11c_3 = 3c - b + a + b + a$$

$$c_1 \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} + c_2 \begin{bmatrix} 2 \\ 1 \\ 3 \end{bmatrix} + c_3 \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$$

$$a = b = c = 0$$

$$c_3 = 0$$

$$c_2 = 0$$

$$c_1 = 0$$

} already independent